of such experiments can only be known to Alice or Bob if they come together and compare notes.

The first question is whether Alice and Bob can simultaneously measure their own observables. We have seen that there are quantities that cannot be simultaneously measured. In particular, two observables that do not commute cannot both be measured without the measurements interfering with each other. But for Alice and Bob, it is easy to see that every component of σ commutes with every component of τ. This is a general fact about tensor products. The operators that act on the two separate factors commute with one another. Therefore, Alice can make any measurement on her spin and Bob can make any measurement on his, without either interfering with the other's experiment.

Let's suppose Alice measures σ_z and Bob measures τ_z, and then they multiply the results. In other words, they conspire to measure the product $\tau_z\sigma_z$.

The product $\tau_z\sigma_z$ is an observable that is mathematically represented by first applying σ_z to a ket and then subsequently applying τ_z. Keep in mind that these are just the mathematical operations that define a new operator: they are different from the act of performing a physical measurement. You don't need an apparatus to multiply two operators; you just need a pencil and paper. Let's see what happens if we apply the product $\tau_z\sigma_z$ to the state $|sing\rangle$:

$$\tau_z\sigma_z\frac{1}{\sqrt{2}}\Big(|ud\rangle - |du\rangle\Big).$$

First, using the table in Eqs. 6.8, apply σ_z:

$$\tau_z\sigma_z\frac{1}{\sqrt{2}}\Big(|ud\rangle - |du\rangle\Big) = \tau_z\frac{1}{\sqrt{2}}\Big(|ud\rangle + |du\rangle\Big).$$

Now, apply τ_z to get